

Design and Simulation of Tunable Transmittance Optical Filter Using Phase Change of Sb_2Se_3

Surawut Wicharn¹, Chettanat Padungatthakij², Suebtarkul Suchat³, and Prathan Buranasiri^{2*}

¹ Department of Physics, Faculty of Science, Srinakharinwirot University, Bangkok, Thailand.

² Department of Physics, School of Science, King Mongkut's Institute of Technology Ladkrabang, Bangkok, Thailand.

³ Department of Physics, Faculty of Science and Technology, Phranakorn Rajabhat University, Bangkok, Thailand.

* prathan.bu@kmitl.ac.th

Abstract:

We proposed a switchable optical filter designs using antimony triselenide (Sb_2Se_3) to operate around 1064-nm wavelength. Sb_2Se_3 is a phase change material (PCM) which has low absorption in the IR spectrum, high and stable refractive index contrast between the two phases from visible through IR spectrum, and a high phase switching optical endurance up to an order of 10^6 cycles. We show the reasoning behind selecting long-pass edge filter design and compare its performance to other planar thin-film designs. We also show the tunability of the design, and increase the optical performances using a simple optimization. (96 words)

Keywords: switchable filter, antimony triselenide, phase change material (keywords < 7)

1. Introduction

Phase change material (PCM) can have different meanings depending on the field. In energy storage, the term would refer to material that store and release thermal energy through changes in its state of matter (latent heat), and the change occurs between solid and liquid states [1]. However, in electronics and photonics, the term PCM refers to a solid-state material that changes its crystal structure when its temperature rise above transition temperature (T_{trans}). This change in crystalline structure leads to a change in its optical properties, e.g. complex refractive index ($\tilde{n}$) which comprised of real refractive index (n) and imaginary extinction coefficient (k) parts [2]. The refractive index contrast between the two phases in PCM is an order of magnitude larger than contrast induced by applied electric field or light intensity (Kerr effect) [3], this property makes PCM ideal for photonic applications that require the modulation of light transmittance, absorptance, or reflectance. Examples include reconfigurable metasurface [2, 4-6], optical limiters [3, 7-11], optical switches [12, 13], optical filters [14-16], and absorbers [17, 18].

Phase change material can be classified by its behavior after phase transition. A volatile PCM will revert to its pre-transitioned phase if temperature falls below T_{trans} . But a non-volatile PCM will remain in its transitioned phase even if temperature falls below T_{trans} . Non-volatile PCM can be turn back to its pre-transitioned phase by increasing the temperature to melting temperature and then quenched at a speed faster than the molecules can organized into crystal structure.

An example of volatile PCM is vanadium dioxide (VO_2). In the past 10 years, VO_2 has gathered attention as a volatile PCM with T_{trans} closest to room temperature at 68 °C, it also has a relatively large refractive index contrast starting from near-IR (NIR), with small extinction coefficient (k) in the insulating phase (cold state), but its k became higher when transitioned to metallic phase (hot state) [19]. These properties make VO_2 an ideal material for optical limiter [7-11], which is a device that is used to protects sensitive detectors or human eyes from high-power laser by limiting the output of high-intensity light (OFF mode), but won't reduce the output of low-intensity light (ON mode). To protect the eyes or sensors, the incident light needs to heat VO_2 until it transitioned to metallic phase at $T_{\text{trans}} = 68$ °C. Attempts have been make to reduce T_{trans} closer to room temperature to reduce time required for phase transition, but these methods reduce the switching endurance and refractive index contrast [20], or required constant power source [11]. Beyond the above mentioned applications, VO_2 has also been apply to thermochromic window coating that helps with temperature regulation inside a room by blocking high-intensity sunlight in the Summer, but allows lower-intensity sunlight in the Winter to heat the room [21, 22].

Examples of non-volatile phase change materials are chalcogenides such as $\text{Ge}_2\text{Sb}_2\text{Te}_5$ (GST) or $\text{Ge}_2\text{Sb}_2\text{Se}_4\text{Te}$ (GSST), with GST being one of the earliest and most studied PCM [23]. GST has large refractive index contrast between amorphous and crystalline phases, low T_{trans} when compared to other non-volatile PCM, $T_{\text{trans}} \sim 145$ °C for cubical phase, and $T_{\text{trans}} \sim 200$ °C for hexagonal phase [3, 24]. GST drawback is its large extinction coefficient in both visible and IR spectra which causes high loss from absorption. Meanwhile, GSST ($\text{Ge}_2\text{Sb}_2\text{Se}_4\text{Te}_1$) is another popular non-volatile PCM, it is a derivative of GST that substitutes some part of tellurium with selenium. It has a large refractive index contrast in the visible spectrum but has a lower extinction coefficient which helps increase optical efficiency. But GSST has a higher T_{trans} at ~ 242 °C [24].

Compared to the other two non-volatile PCMs, antimony triselenide (Sb_2Se_3) is a relatively new material in the field of photonics, **only one review paper related to photonic applications of Sb_2Se_3 existed** [25]. **(recheck claims whether it's still true)**. The initial research on Sb_2Se_3 is in solar-energy applications, but Sb_2Se_3 has recently gathered interest as a non-volatile PCM for photonic applications because of its near-zero extinction coefficient in the near-IR and IR spectra, its extinction coefficients are also lower than GST in the visible spectrum for both amorphous and crystalline phases, it has a high and stable refractive index contrast from visible to infrared spectra, and the transition temperature of Sb_2Se_3 is between GST and GSST at $\sim 186^\circ\text{C}$ [25, 26]. Sb_2Se_3 has a fast switching speed in a microsecond ranges [23, 26, 27]. Its phase-switching endurance is also high, with experimental demonstrations showing that it can reach as high as 10^6 to 10^7 cycles with an optimized laser schemes for phase transition [26, 27]. These properties make Sb_2Se_3 an attractive material for low-loss, reconfigurable optical switch or optical circuit that are used in optical waveguide [4-6, 13, 23].

Similar to optical limiter, switchable optical filter also operates in two-mode, where the intensity of incident light can be reduced depending on the current mode. In optical limiter case, the mode changing mechanism is self-activated, i.e., PCM needs to undergoes phase transition by the absorption from incident light [9, 10]. This is because the switching time from ON to OFF mode needs to be minimized so that as little high-intensity light can reaches the eyes or detector as possible. But for optical switch, the switching time is not the main requirement and the switching mechanism can be from other mean such as joule heating [15], or light-induced heating using laser with wavelength outside of its intended operating bandwidth (e.g., if the device is intended to operates at IR wavelength, the phase transition can be activated by visible laser) [26]. Both optical filter and optical limiter required maximize transmittance and zero reflectance in ON mode, and minimize transmittance in OFF mode for the best efficiency. But optical limiter further required absorptance in ON mode as a self-activating switching mechanism. The availability of these two switching mechanisms in switchable filter is beneficial because it eliminates the required absorptance in the operation bandwidth for inducing phase transition, which helps increase the efficiency of the design. The designs of these two type of devices can be a planar thin-film structure or a metasurfaces; planar structures are easier to design, optimize and fabricate, but metasurface structures offer more design flexibility [10]. For a planar designs, there are many methods to create high transmission and reduce reflectance, such as: 1) single or multi-layers antireflection designs and 2) Line-pass, long-pass, or short-pass filters [28].

In this work, we investigate Sb_2Se_3 as a material for switchable filter or optical limiter in a planar thin-film designs. The intended operating wavelengths are selected to be 1064 nm because they are from Nd:YAG laser which are commonly used in laboratory setups. At this wavelength, Sb_2Se_3 has zero absorption in amorphous phase and near-zero in crystalline phase. Its refractive index contrast is stable throughout the IR spectrum, which should enable a broadband design. The goal is to create a broadband optical limiter that have high transmission in the ON mode and reflected most of the incident light in OFF mode.

2. Theories

2.1 Optical properties of Sb_2Se_3

Antimony triselenide (Sb_2Se_3) is an orthorhombic chalcogenide non-volatile phase change material (PCM) characterized by quasi-one-dimensional ribbons. While its amorphous state consists of a disordered network of distorted motifs, crystallization restores long-range order along the ribbon direction. This thermally driven transition can be triggered by nanosecond to microsecond optical or electrical pulses [23, 26]. This reversible process supports high endurance, with optical switching reaching 10^6 to 10^7 cycles. As an optical device, the material enables repeatable toggling between high and low transmission modes, provided the local heating stays below the melting or ablation threshold [4, 27].

The complex refractive index ($\tilde{n}$) of any phase change material at any wavelength λ can be written as:

$$\tilde{n}_p(\lambda) = n_p(\lambda) + ik_p(\lambda) \quad (1)$$

where the real term n is the refractive index, the imaginary term k is extinction coefficient, and p denotes the phase which for Sb_2Se_3 can either be amorphous (*amo*) or crystalline (*cry*). Dispersive models, such as Tauc-Lorentz or multi-oscillator fits are often used to represent the measured n_p and k_p over the spectral range of interest. But in this work, a linear interpolation is used since all the materials dataset used have the measurement interval of less than 4 nm. The complex refractive index ($\tilde{n}$) at 1064 nm are: $\tilde{n}_{amo} = 3.370 + 0.000i$, and $\tilde{n}_{cry} = 4.308 + 0.010i$. Figure 1 shows their $\tilde{n}$ from 300 to 1700 nm using dataset from Delaney, et al. [23]. Blue lines represent real refractive index (n), and red lines represent imaginary extinction coefficient (k). In amorphous phase (Fig. 1a), the refractive index increases from ~ 2.4 at 300 nm to a peak of ~ 4.0 at 600 nm and gradually decreases to a stable value of ~ 3.3 above 1200 nm, indicating a strong dispersion near the band edge of visible spectrum and a weaker dispersion in the NIR spectrum. The extinction coefficient starts from ~ 2.1 at 300 nm and dropped to zero at around 850 nm. In the crystalline phase (Fig. 1b), the refractive index at the UV wavelengths has around the same value as that of amorphous phase but reached a higher peak of ~ 5.1 at around 660 nm, it is then dropped to a stable

of ~ 4.1 in the NIR spectrum. The k -index of crystalline phase peaked at around 3.1 in the near-UV spectrum and rapidly approach zero above 900 nm.

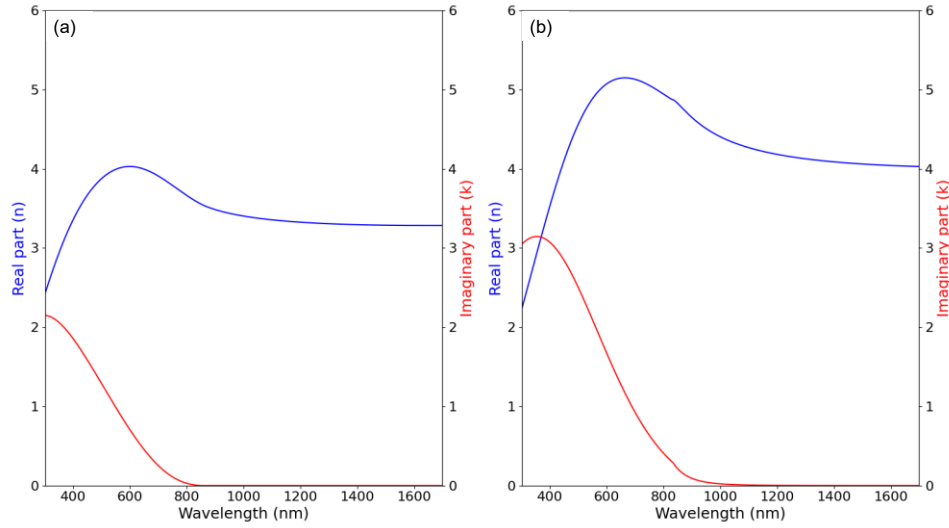


Fig. 1. Complex refractive index of Sb_2Se_3 in (a) amorphous, and (b) crystalline phases. The graphs are plotted from experimental data from Delaney, et al. [23].

The refractive index contrasts ($\Delta n = n_c - n_a$) are large and stable from the visible through IR spectra, and the extinction coefficients are large in the near-UV but falls to zero around NIR spectrum for both phases, with crystalline phase exhibits higher absorption in the visible spectrum. So, when compared to other PCMs, Sb_2Se_3 has several advantages when operating in NIR and IR wavelengths. The large Δn and low k starting from NIR wavelengths enable a device to operate in two modes with little to no optical loss, stable Δn also helps with broadband designs.

2.2 Single-layer antireflection design

In designing optical devices, most devices will either be optimized for antireflection (AR) or high reflection properties. But for two-mode optical devices, i.e. optical filter and optical limiter, both ON and OFF modes do not utilize reflectance for any purpose, so the design will be antireflection. We will first consider its reflectance in ON mode, and then consider the reflectance contrast (ΔT) between ON and OFF mode.

The simplest antireflection thin-film device is a single film layer deposited on a substrate; both the film and substrate should have no absorptance. A single-layer device reflects incident light at two interfaces as shown in Fig. 2: the bottom interface is between thin film and substrate (R_1), and the top interface is between air and film (R_2). Antireflection condition occurs when destructive interference between R_1 and R_2 happens, this requires same amplitude and a phase difference of π radian (optical path length of $\lambda/2$). Since R_1 propagated through thin film layer twice (while transmitted down and reflected backup), the film optical thickness is equal to half of optical path, an optical thickness of $\lambda/4$ which is also known as quarter-wave optical thickness (QWOT). Optical thickness (OT) is related to the physical thickness (t) by:

$$OT(\lambda) = n(\lambda)t, \quad (2)$$

where $n(\lambda)$ is the refractive index of the film at λ . For a single-layer AR device, the thin-film physical thickness (t) needs to be equal to:

$$t = \frac{OT}{n} = \frac{\lambda/4}{n}. \quad (3)$$

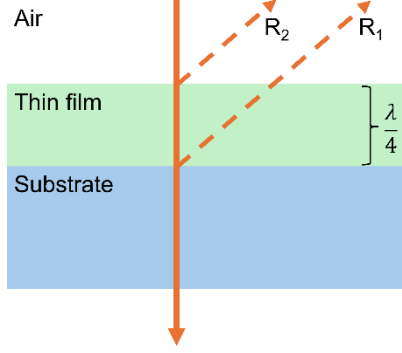


Fig. 2. Reflected light paths of a single-layer thin film device. R_1 is the reflected light at the film-substrate interface, and R_2 is the reflected light at the air-film interface. Antireflection occurs when the optical path difference between the two is $\lambda/2$. The reflection angles are exaggerated for illustration purpose.

Another method to calculate the thickness required for AR condition is to consider the reflection of the device. The reflectance (R) of just the substrate is related to its Fresnel coefficient (r) by:

$$R = |r|^2 = \left| \frac{n_a - n_s}{n_a + n_s} \right|^2, \quad (4)$$

where n_a is air refractive index, and n_s is substrate refractive index. If we add a single-layer film on top of a substrate, film-substrate layers can combine to a so called effective reflectance index (n_r) [28], defined as:

$$n_r = n_f \frac{(n_s + n_f) + (n_s - n_f)e^{-i2\theta}}{(n_s + n_f) - (n_s - n_f)e^{-i2\theta}}, \quad (5)$$

where n_f is film refractive index, i denotes imaginary unit, and θ is phase thickness which is related to optical thickness (OT) and physical thickness (t) of the film by:

$$\theta = k_0 OT = k_0 nt, \quad (6)$$

where k_0 is the angular wavenumber ($k_0 = 2\pi/\lambda$). By replacing n_s in Eq. 4 with n_r from Eq. 5, the reflectance of a single-layer device become:

$$R = |r|^2 = \left| \frac{n_a - n_r}{n_a + n_r} \right|^2. \quad (7)$$

Antireflection condition occurs when $R = 0$ or $n_r = n_a$. This condition occurs when the phase thickness (θ) in Eq. 5 is in multiple of $\pi/2$ radian:

$$\theta = Z \frac{\pi}{2}, \quad (8)$$

where Z can be any even integer. If this condition is satisfied, Eq. 5 can be simplified using Euler's identity ($\exp^{i\pi} = -1$) to:

$$n_r = n_f \frac{(n_s + n_f) - (n_s - n_f)}{(n_s + n_f) + (n_s - n_f)} = \frac{n_f^2}{n_s}, \quad (9)$$

By replacing n_r from Eq. 7 with Eq. 9, an antireflection condition happens when $n_r = n_a$, or:

$$n_a n_s = n_f^2. \quad (10)$$

Another condition that needs to be considered is the relative refractive index between air, film, and substrate, which must be in an order of $n_{air} < n_f < n_{substrate}$ since light shift its phase by π radian when reflected from lower to higher refractive index medium. The reflected light (R_2) at air-film interface always undergoes π radian phase shift since n_{air} will be the lowest in most cases, to ensure destructive interference, R_1 must also phase shifted by the same value.

These two conditions place a limitation on material selection, because there exist only one refractive index answer for each n_s for a complete antireflection property. So, the QWOT design must have the film refractive index higher than air but lower than substrate (silica; SiO_2), but Sb_2Se_3 has refractive index higher than silica at any wavelengths. The scenario from QWOT design now become; R_1 travelled 2QWOT (π radian) further than R_2 but only R_2 has its phase shifted by π radian during reflection at the air-film layer, so a constructive interference occurs instead. To restore destructive interference, the film needs to be $\lambda/2$ or half-wave optical thick (HWOT) instead so that R_1 will travel additional 2HWOT (2π radian) further than R_2 . As an example, at the reference wavelength $\lambda_0 = 1064$ nm, if we use 3-mm-thick SiO_2 with a refractive index of $n_s = 1.46$ as substrate [29], the ideal QWOT-film thickness is calculated from Eq.10 to a refractive index of 1.21 for a complete antireflection design. Since refractive index of 1.21 is lower than any commonly available material, a material with the closest refractive index is MgF_2 at $n_{\text{MgF}_2} = 1.38$ [30], this difference cause the reflection to be higher than an ideal film as illustrated in Fig. 3. In the case of Sb_2Se_3 (blue), its thickness changes to HWOT, but its lowest reflectance value is still higher than MgF_2 since the $n_{\text{Sb}_2\text{Se}_3} = 3.37$, much larger difference to an ideal value than MgF_2 . An ideal film also exhibits some reflection because we also modelled the reflection from the substrate-air interface on the other side.

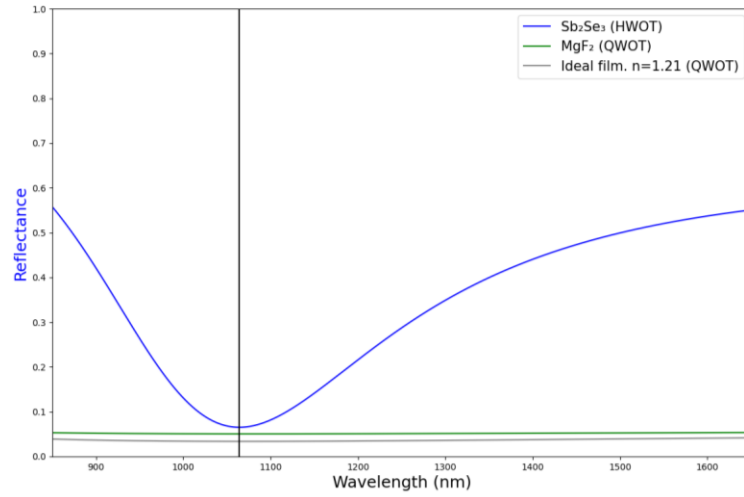


Fig. 3. Reflectance comparison between each single-layer thin film designs. At 1064 nm (black vertical line), Sb_2Se_3 (blue) has reflectance $R = 6.5\%$. MgF_2 with $n_{\text{MgF}_2} = 1.38$ (green) has $R = 5.0\%$. And an ideal film with $n_f = 1.21$ (grey) has $R = 3.4\%$. The thickness of Sb_2Se_3 is HWOT, both ideal film and MgF_2 are QWOT thick.

2.3 Multilayer antireflection design

A two-layer device with QWOT in each layer (Q/Q design) can also be used to create antireflection condition. The Fresnel reflection coefficient (r) from Eq. 7 can be expanded to:

$$r = \frac{n_a - n_{r2}}{n_a + n_{r2}} = \frac{n_a - \left(\frac{n_{f2}^2}{n_{r1}}\right)}{n_a + \left(\frac{n_{f2}^2}{n_{r1}}\right)} = \frac{n_a - \left(\frac{n_{f2}^2}{n_{f1}/n_s}\right)}{n_a + \left(\frac{n_{f2}^2}{n_{f1}/n_s}\right)}, \quad (11)$$

where the n_f numbering begins from substrate side and increases toward air ($n_s / n_{f1} / n_{f2} / n_a$), n_{r2} is the effective reflectance index for the entire structure [28], and n_{r1} is the effective reflectance index of only substrate (n_s) and bottom film (n_{f1}). Similar to Eq.7 antireflection condition occurs when:

$$n_a = \frac{n_{f2}^2}{n_{f1}/n_s}, \quad (12)$$

or:

$$\frac{n_a}{n_s} = \frac{n_{f2}^2}{n_{f1}^2}. \quad (13)$$

This means that the refractive index of the top film must be smaller than the bottom film ($n_{f2} < n_{f1}$). As an example, at 1064 nm if we use the same SiO_2 as substrate ($n_s = 1.46$) and Sb_2Se_3 as bottom film ($n_{f1} = 3.37$), the required refractive index of n_{f2} can be calculated to 2.78. A commonly used material with the closest refractive index is TiO_2 with $n_{f2} = 2.31$ [31]. When compared to a single-layer Sb_2Se_3 design, two-layer design allows for wider materials selection, has lower reflectivity, and broader operation wavelength, as shown in Fig. 4. Another way to create two-layer AR condition is to fix the two film refractive index and vary each film thickness instead, this is explained in chapter 5.2 of [28]. However, this method narrows the antireflection bandwidth.

A three-layer design (Q/H/Q) has a HWOT layer placed in the middle. The required refractive index orderings of each layer are $n_s < n_{f1} < n_{f2}$ and $n_{f2} > n_{f3} > n_a$. The middle layer (n_{f2}) acts as an absentee layer when incident wavelength (λ) is the same as reference wavelength (λ_0), but it acts as compensating layer to restore resonance for a scenario where λ differs from λ_0 , a more detailed explanation is provided in chapter 5.4 of [28]. The Fresnel reflection coefficient from Eq. 7 is equal to:

$$r = \frac{n_a - n_{r3}}{n_a + n_{r3}} = \frac{n_a - \left(\frac{n_{f3}^2}{n_{r2}}\right)}{n_a + \left(\frac{n_{f3}^2}{n_{r2}}\right)}. \quad (14)$$

Because the middle film layer (n_{f2}) is HWOT thick, it behaves as an absentee layer [28, 32]. So $n_{r2} = n_{r1}$ and Eq. 14 becomes:

$$r = \frac{n_a - \left(\frac{n_{f3}^2}{n_{r1}}\right)}{n_a + \left(\frac{n_{f3}^2}{n_{r1}}\right)} = \frac{n_a - \left(\frac{n_{f3}^2}{n_{f1}/n_s}\right)}{n_a + \left(\frac{n_{f3}^2}{n_{f1}/n_s}\right)}, \quad (15)$$

so, a three-layer antireflection condition occurs when:

$$\frac{n_a}{n_s} = \frac{n_{f3}^2}{n_{f1}^2}. \quad (16)$$

The refractive index ratio between top and bottom film is similar to a two-layer design (Eq. 13). As example; using Eq. 16, if the bottom layer (n_{f1}) is Sb_2Se_3 (QWOT) then the top layer (n_{f3}) refractive index is calculated to be 2.78, so TiO_2 ($n_{\text{TiO}_2} = 2.31$) is used again. The middle layer needs to have the highest refractive index, so Silicon (Si) with $n_{\text{Si}} = n_{f2} = 3.55$ [33] is used. As shown in Fig.4, this three-layer design reflectance at wavelength further from λ_0 (i.e., 1400 nm and longer) is lower than that of a single or two-layer design. This is the benefit of adding Si as middle layer, as its contribution to a broader operational bandwidth [32].

Instead of placing PCM as the bottom film layer, Sb_2Se_3 can be moved to the middle layer so that the selection of ideal n_{f1} and n_{f3} becomes easier. The bottom layer could be Si_3N_4 ($n_{f1} = 2.00$) [34] so that the ideal top layer calculated from Eq. 16 have $n_{f3} = 1.65$, which Al_2O_3 has the closest refractive index $n_{\text{Al}_2\text{O}_3} = 1.62$ [31]. These design are shown in Fig. 4 where the reflectance of the single-layer, two-layer, and three-layer designs, which have Sb_2Se_3 placed directly on top of the substrate ($n_{f1} = \text{Sb}_2\text{Se}_3$), are 6.5% to 6.7%. But the last design with Sb_2Se_3 placed in the middle ($n_{f2} = \text{Sb}_2\text{Se}_3$) has reflectance equal to 3.4%, the same value as an ideal single-layer design (3.4%) in Fig. 2. This improvement can be explained by the selection of both n_{f1} and n_{f3} that are close to ideal values.

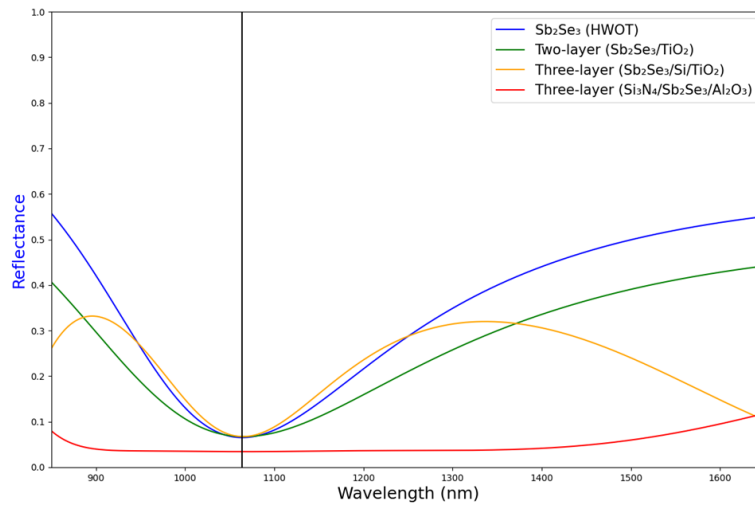


Fig. 4. Reflectance comparison between a single-layer and multi-layer Sb_2Se_3 designs. At 1064nm (black vertical line), the single-layer, two-layer and three-layer designs with Sb_2Se_3 as bottom layer (n_{f1}) have similar reflectance at 6.5 - 6.7%. The three-layer design with Sb_2Se_3 as middle layer (n_{f2}) has the lowest reflectance at 3.4%.

2.4 Long-pass edge-filter design

In all previous designs, the optical performance after phase transition has not been considered yet. Aside from low reflectance, another requirement for two-mode optical device is high optical contrast between the two modes. Figure 5 shows the reflectance comparison between Sb_2Se_3 in amorphous (solid) and crystalline (dotted) phases, which represents the device in ON and OFF modes, respectively. The designs are the same as described in Fig. 4. The single-layer design has the highest reflectance contrast of 55.0% (Fig. 5a, blue). For the three-layer design with Sb_2Se_3 as the middle layer (Fig. 5b, red), even though it has the lowest reflectance in amorphous phase among other designs at 3.4%, its crystalline phase reflectance is only 14.8%, which make the reflectance contrast of only 11.4%.

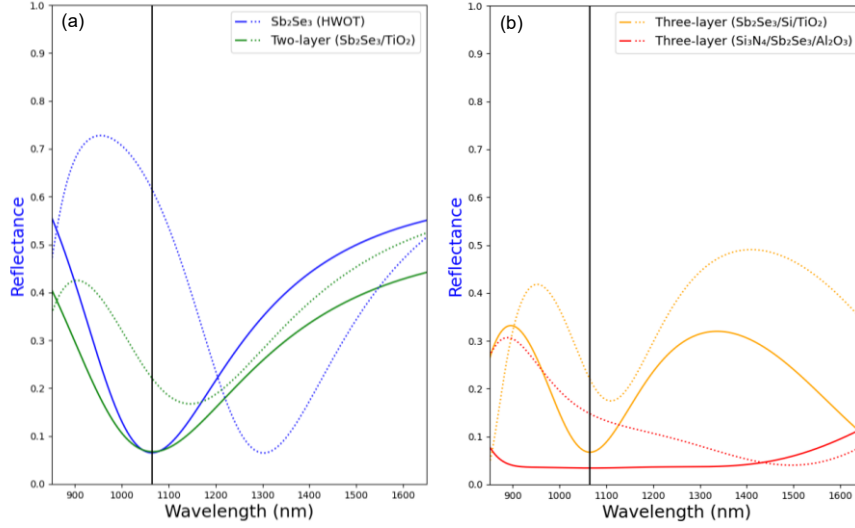


Fig. 5. Reflectance comparison between designs shown in Fig. 4 before (ON mode, solid) and after (OFF mode, dotted) mode switching. (a) Single-layer (blue) and two-layer designs (green). (b) Three-layer designs with Sb_2Se_3 as bottom layer (orange) design with Sb_2Se_3 as middle layer (red). Single-layer design has the highest reflectance contrast. The vertical black lines indicate $\lambda_0 = 1064$ nm.

Edge-filter design has high reflection in a region around reference wavelength (λ_0) but exhibits strong antireflection (AR) property outside of this region. The device could be designing such the AR area at the edge outside of high-reflection region is at the desired operating wavelength. And when the device changes its mode to OFF (phase change material transitioned to crystalline phase) the reflectance profile will be shifted, and the area around the operating wavelength would be in high-reflection area instead. Edge filter design consists of QWOT-thick high refractive index (H) and low refractive index (L) layers deposited on top of each other symmetrically in a structure called unit cell (U), the number of unit cells could be one or more. The one side of the AR areas outside the high-reflection region will be stable depending on the structure of unit cell; a long-pass structure ($\frac{H}{2}L\frac{H}{2}$) has a stable AR area in longer wavelengths, and the short-pass structure ($\frac{L}{2}H\frac{L}{2}$) has a stable AR area in a shorter wavelength, where $1/2$ denotes a layer with half-QWOT thickness. To minimize the thickness of the device, a long-pass design is preferred since λ_0 will be at a shorter wavelength. The spectral width of high-reflection region is directly correlated to the refractive index differences between H and L layers, and the steepness of the region's edges is directly correlated to the number of unit cells.

Figure 6 shows five long-pass designs ($\frac{H}{2}L\frac{H}{2}$) with number of unit cells (N) ranging from 1 to 8. These designs use Sb_2Se_3 as H layer and MgF_2 as L layer, the reference wavelength $\lambda_0 = 740$ nm, and are deposited on the same 3-mm-thick silica substrate used on previous examples. When devices are in ON mode or amorphous phase (Fig. 6a) the area around the interested wavelength ($\lambda = 1064$ nm) is in antireflection region, but when devices switch to OFF mode or crystalline phase (Fig. 6b) the area around interested wavelength is in high-reflection region. These create a high reflectance contrasts between the two modes of more than 90% for 4, 6 and 8 unit-cell designs, much higher than any previously shown designs. It can also be seen that increasing number of unit cells make the region edge steeper but the improvement from 6 to 8 unit cells is lower than from 1 to 2 unit cells, so the benefits of increasing edge sharpness should be considered with the cons of increasing device thickness.

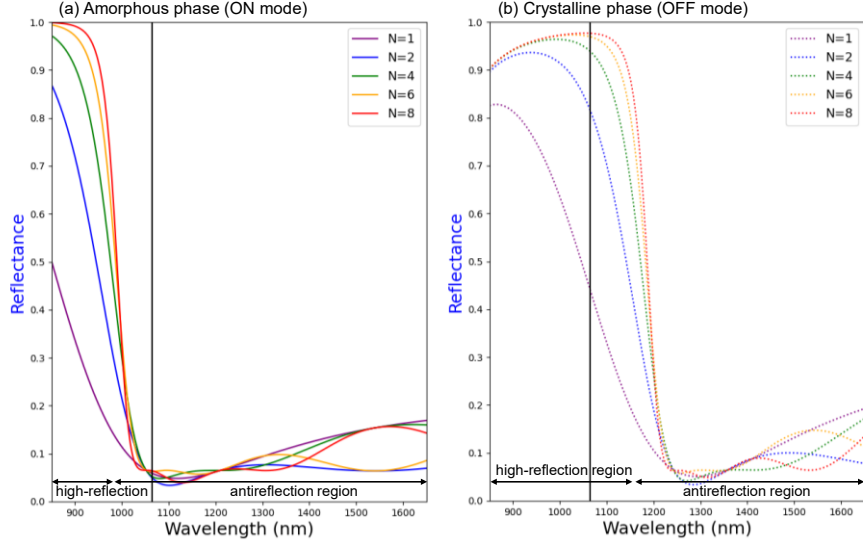


Fig. 6. Reflectance comparison of long-pass edge-filter designs ($\frac{H}{2}L\frac{H}{2}$) with varying number of unit cells. (a) When devices are in amorphous phase (ON mode), the interested wavelength (black vertical line) is in AR region. (b) When devices are in crystalline phase (OFF mode), the interested wavelength is in high-reflection region. Reference wavelength $\lambda_0 = 740$ nm.

3. Methods

3.1 Simulation of optical performances

In this work, we use transfer matrix method (TMM) of Fresnel equation to calculate the transmittance, absorptance and reflectance of every designs, the simulation is largely based on ‘Optical Thin Film Design’ [28], **although a simulation software such as COMSOL would also be applicable.** In some simulation works, the substrate index is considered to be equal to air ($n_{sub} = n_{air} = 1.0$) to simplify the design and allows for a more widely applicable analysis methods. But in this work we use 3-mm thick silica (SiO_2) as substrate ($n_s = 1.47$) [29], and calculate the backside reflection from substrate-air layer on the other side. The high-index layers (H) are Si [35], TiO_2 [31], or Si_3N_4 [34]. Their refractive indices at 1064 nm are 3.56, 2.31, and 2.00, respectively. The low-index layers (L) are Al_2O_3 [31], SiO_2 or MgF_2 [30]. Their refractive indices are 1.62, 1.45, and 1.38, respectively. All the above mentioned refractive index dataset can be found in refractiveindex.info website [36]. But because antimony triselenide (Sb_2Se_3) is a relatively new material in this field, its dataset cannot be found there. We use Sb_2Se_3 dataset from Delaney, et al. [23], as shown in Fig. 1. The refractive index of air is set to $n_a = 1.0$ across the spectrum. The incident angle is 0° and light is transverse electric (TE) polarized unless stated otherwise.

Our devices are designed to operate at the operating wavelength center around 1064 nm. The design comparison will be between a single-layer, two-layers, three-layers, and edge-filter designs. The goal is to design a device with high transmission contrast (ΔT) between the two phases of Sb_2Se_3 by minimizing the reflectance in ON mode, and maximize R in OFF mode.

3.2 Simulation of optical endurance and stability under large phase switching cycles

Experimental studies have demonstrated that Sb_2Se_3 can have phase switching cycles with minimal degradation to its optical contrast up, and maintain a low-loss, reversible modulation in an order of 10^6 cycles [26]. Even so, Sb_2Se_3 still exhibits slight complex refractive index drift due to structural relaxation and cumulative thermal damage. Accounting for these variations is essential for ensuring long-term reliability and precision in high-endurance photonic applications [26, 27]. To account for this, the complex refractive index ($\tilde{n}$) can be rewritten from Eq. 1 as:

$$\tilde{n}_p(C) = n_p(C) + ik_p(C), \quad (17)$$

where C is the number of phase switching cycle.

The real part of the refractive index (n) is described by an exponential relaxation towards a saturated value such that:

$$n_p(C) = n_{p,\infty} - (n_{p,\infty} - n_{p,0})\exp^{-\alpha_p C}, \quad (18)$$

where $n_{p,\infty}$ is the asymptotic index after many cycles, $n_{p,0}$ is the refractive index at the first cycle (obtained from ellipsometry or literature data), and α_p is the drift coefficient per cycle of phase p [23, 37].

Meanwhile, the imaginary part (k) can be approximated from a linear growth as:

$$k_p(C) = k_{p,0}(1 + \beta_p C), \quad (19)$$

where $k_{p,0}$ is the initial loss and β_p is the loss growth coefficient. The parameter β_p is the relative loss per cycle of phase p [26, 38]. These models from Eq. (XXA) and (XXB) are adopted in this work, they are consistent with experimental trends and with general PCM endurance theory, providing a practical way to assess contrast degradation and threshold drift of PCM-based optical limiters over large cycles [23, 26, 37, 38].

The endurance of optical limiters can be quantified using the following metrics: 1) transmitting contrast degradation, and 2) insertion loss.

Transmittance Contrast Degradation:

Degradation is defined as the reduction in transmittance contrast after N cycles, as in:

$$D(C) = \frac{\Delta T(C) - \Delta T_0}{\Delta T_0}, \quad (20)$$

where transmittance contrast ($\Delta T_0 = T_a - T_c$) is the difference between amorphous (ON) and crystalline (OFF) phases at the initial cycle ($C = 0$).

Insertion loss:

The insertion loss (IL) of the device at cycle C is the ratio between the transmittance at C cycle and a reference state of cycle 0 [39]:

$$IL_{dB}(C) = 10 \log \left[\frac{T_c(C)}{T_c(0)} \right]. \quad (21)$$

A growing $IL(N)$ indicates increasing loss, which is typically driven by the linear growth of the imaginary index $k_p(C)$ and by detuning of the resonance due to drift of $n_p(C)$.

4. Results & Discussions

4.1 Optical properties of 1064-nm optical limiters

The design architectures, number of layers, Sb_2Se_3 required for each design, overall device thickness, transmittance in each mode, and transmittance contrast (ΔT) of each design are shown in Table 1. The interested operating wavelength is at 1064 nm. The reference wavelength (λ_0) is 1064 nm for single-layer, two-layers, and three-layers designs, but λ_0 is set to 740 nm for all edge-filter designs. Only single-layer design and edge-filter designs with at least two unit cells ($N \geq 2$) have a higher ΔT than 50%, with 7-unit-cells design having the highest $\Delta T = 95.1\%$. It's important to note that there is only marginal improvement in optical performances when increasing the number of unit cells above 3. For examples; ΔT improved by 2% in $N = 4$ design when compared to $N = 3$ design, but the thickness increased by 33.3%, and when comparing $N = 5$ to $N = 4$ designs, ΔT increased by only 0.6% but the device needs to be 25.0% thicker. This benefits and drawback are an important consideration, minimizing device thickness also has the benefit of reducing the energy required for phase transition since the total Sb_2Se_3 used will be thinner. In edge-filter designs, even though there are always some reflectance in OFF mode (Fig. 6b), the transmittance in OFF mode decreases with increasing number of N , with $N \geq 6$ have 0% transmittance. This is because N correlated to the total Sb_2Se_3 thickness and the extinction coefficient of Sb_2Se_3 in crystalline phase is small but not zero ($k_{\text{Sb}_2\text{Se}_3, \text{cry}} = 0.0097$), so they will still be some absorption. For reference, this is still an order of 10^2 smaller than other PCMs such as GST ($k_{\text{GST}, \text{cry}} = 2.4540$) or VO_2 ($k_{\text{VO}_2, \text{metallic}} = 1.4580$).

Table 1. Designs of 1064-nm Sb_2Se_3 -based optical limiter. Transmittance is calculated at the interested $\lambda = 1064$ nm. Single, two, and three-layer designs all have $\lambda_0 = 1064$ nm, but all edge-filter designs have $\lambda_0 = 740$ nm.

Design	Layout	Number of layers	SbSe used (nm)	Device size (nm)	T (ON) (%)	T (OFF) (%)	ΔT (%)
Single-layer	SbSe	1	157.9	157.9	93.5	37.2	56.3
Two-layers	SbSe/TiO	2	78.9	194.1	93.3	76.9	16.4
Three-layers	SbSe/MgF/TiO	3	78.9	580.3	93.3	76.9	16.4
Three-layers	SiN/SbSe/AlO	3	157.9	455.7	96.6	83.3	13.3
Edge filter	$(\frac{\text{SbSe}}{2}/\text{MgF}/\frac{\text{SbSe}}{2})^1$	3	48.5	182.5	94.1	54.3	39.8
Edge filter	$(\frac{\text{SbSe}}{2}/\text{MgF}/\frac{\text{SbSe}}{2})^2$	5	97.0	365.0	95.0	16.3	78.7
Edge filter	$(\frac{\text{SbSe}}{2}/\text{MgF}/\frac{\text{SbSe}}{2})^3$	7	145.5	547.6	94.8	4.0	90.8
Edge filter	$(\frac{\text{SbSe}}{2}/\text{MgF}/\frac{\text{SbSe}}{2})^4$	9	194.0	730.1	93.7	0.9	92.8

Edge filter	$(\frac{\text{SbSe}}{2}/\text{MgF}/\frac{\text{SbSe}}{2})^5$	11	242.6	912.6	93.6	0.2	93.4
Edge filter	$(\frac{\text{SbSe}}{2}/\text{MgF}/\frac{\text{SbSe}}{2})^6$	13	291.1	1095.1	94.5	0.0	94.5
Edge filter	$(\frac{\text{SbSe}}{2}/\text{MgF}/\frac{\text{SbSe}}{2})^7$	15	339.6	1277.7	95.1	0.0	95.1
Edge filter	$(\frac{\text{SbSe}}{2}/\text{MgF}/\frac{\text{SbSe}}{2})^8$	17	388.1	1460.2	94.4	0.0	94.4

Figure 7 compares the transmittance contrast (ΔT) of designs in Table 1, a single-layer, and edge-filter designs with unit cell numbers ranging from 1 to 6. Even though edge-filter design with unit cell numbers ranging from 3 to 6 (in increasing order: cyan, green, yellow, and orange) have similar transmittance contrast at the operating wavelength (black vertical line), when we take a look at the transmittance contrast across the spectrum, the increase in N also broaden the spectral width of high ΔT region. This effect should be considered if the device is intended to be operated in a broad spectrum.

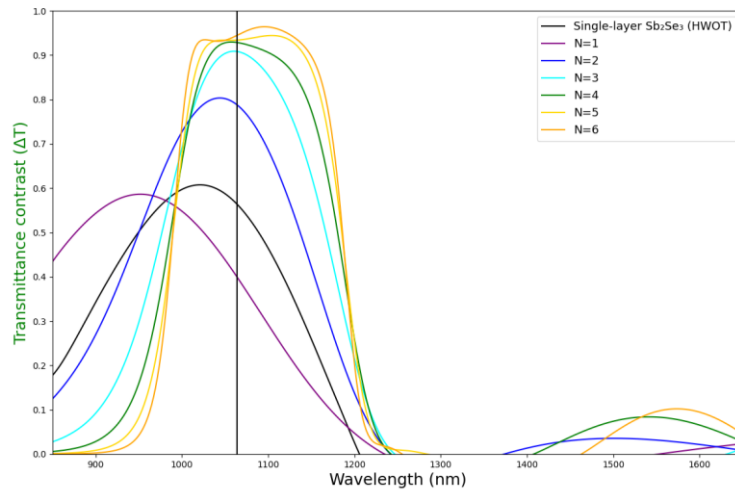


Fig. 7. Transmittance contrast comparison of the design shown in Table 1. The reference wavelength (λ_0) of the single-layer design is 1064 nm. All edge-filters designs have $\lambda_0 = 750$ nm.

Another two criteria that might be an important considerations are the optical performances at different incident angles, and polarization mode (TE or TM). Figure 8 shows the transmittance contrast of a 4-unit-cells and 5-unit-cells edge-filter designs with incident angles ranging from 0 to 90 degrees, and with either TE or TM polarized. For TE polarized light (Fig. 8a and 8c) the ΔT remains higher than 80% for up to $\sim 45^\circ$. And for TM polarized light (Fig. 8b and 8d) the spectral width of high ΔT region blue-shifted stronger and starts at lower angle than TE polarized light. And when comparing 4-unit-cells to 5-unit-cells designs, the spectral width and shifting angle are both comparable. But the 5-unit-cells design has a higher ΔT at the edges for every incident angles, indicating a better broadband, wide-angle optical devices.

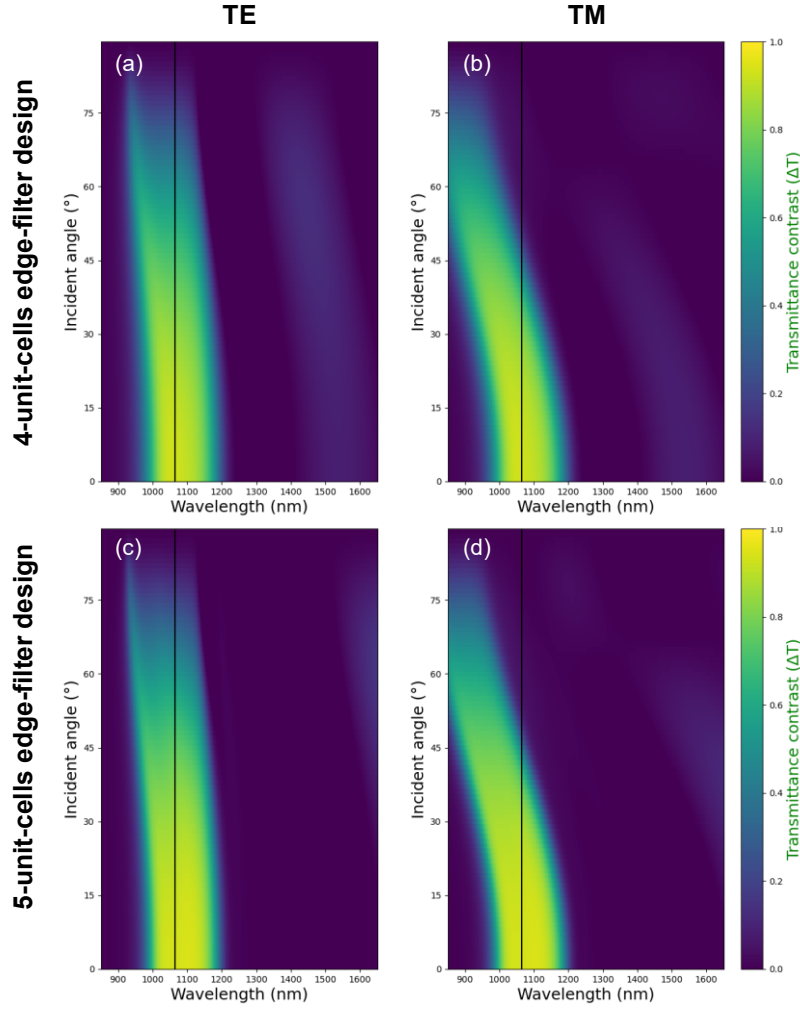


Fig. 8. Heatmaps showing the transmittance contrast at varying incident angles and wavelengths of the 4-unit-cell edge-filter design shown in Table 1. The polarizations of incident light are (a) TE and (b) TM.

With the above mentioned considerations, we decided that the 5-unit-cells long-pass edge-filter design (Fig. 7, yellow) has the best balance between optical performances, device size and fabrication complexity for a broadband, wide-angle configuration.

4.2 Endurance simulation of 1064-nm optical limiters

check endurance of edge-filter design.

In this section, the comparison will be between a single-layer Sb_2Se_3 design, two-layer design ($\text{Sb}_2\text{Se}_3/\text{TiO}_2$), and the long-pass edge-filter design ($N=5$). Figure 9 shows the transmittance of the device in amorphous or crystalline phases at phase switching cycle (C). For the single-layer design (Fig. 9a), the amorphous-phase transmittance stays high and nearly constant over most of the cycling window, whereas the crystalline-phase transmittance slowly decreases with cycling number, indicating a reduction of effective phase contrast in the film. In the two-layers design (Fig. 9b), the addition of TiO_2 layer slightly modifies the absolute transmittance levels but preserves the same qualitative trend of stable high-transmittance amorphous phase and slowly brightening crystalline phase. For the 5-unit-cells long-pass edge-filter design (Fig. 9c), the transmittance of crystalline phase increase over cycles instead, and both states show more pronounced evolution because the resonance design is more sensitive to small refractive index shift, yet the device still maintains a clear separation between amorphous and crystalline transmission after enduring 1.5×10^6 phase switching cycles.

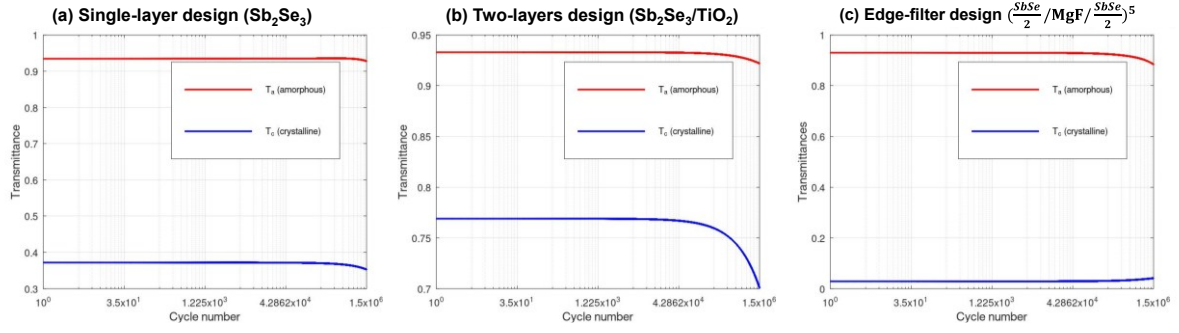


Fig. 9. Transmittance simulation after 1.5×10^6 phase switching cycles in the amorphous (red) and crystalline phases (blue) of (a) single-layer, (b) two-layer, and (c) 5-unit-cells edge-filter devices described in Table 1. **Note the different in y-axis scales of the three subfigures.**

Figure 10 shows the transmittance contrast degradation of the same three designs; the degradation is expressed as a percentage relative to the initial cycle ($C = 0$). Surprisingly, both single-layer and two-layers designs show positive transmittance contrast degradation percentages, especially as the simulations approach the end. Which indicate the modulating performances of these two designs improve with phase switching cycles. This is opposite of edge-filter design, where the contrast degradation is negative, so the modulating performance decrease with usage. The two-layers and edge-filter designs display much larger percentage degradations than the single-layer design, indicates that while resonant enhancement improves low-cycle performance, it also amplifies the impact of long-term drift in phase-change materials.

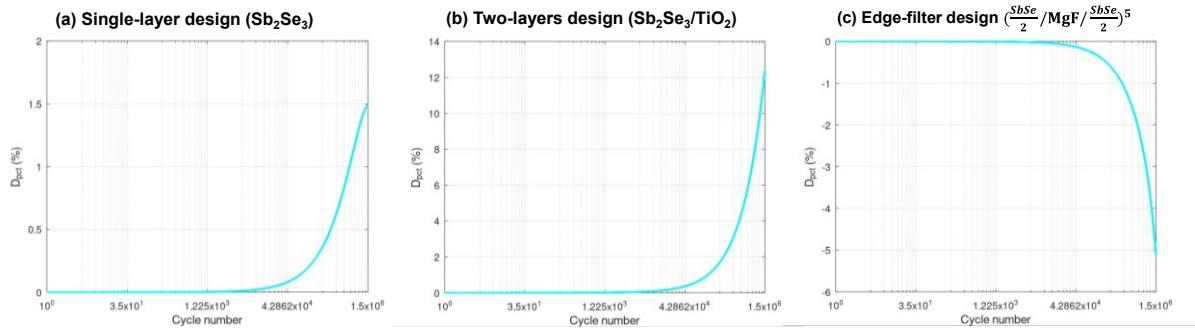


Fig. 10. Transmittance contrast degradation after 1.5×10^6 phase switching cycles of (a) single-layer, (b) two-layer, and (c) 5-unit-cells edge-filter devices. **Note the different in y-axis scales of the three subfigures.**

The insertion losses in the high-transmission amorphous phase of the three designs as a function of switching cycle number is shown in Fig. 11. The insertion losses increase gradually with cycling number for every designs, this reflects the small but cumulative rise in absorption and scattering within Sb_2Se_3 . However, the overall insertion loss after enduring 1.5×10^6 cycles remain within an acceptable limits.

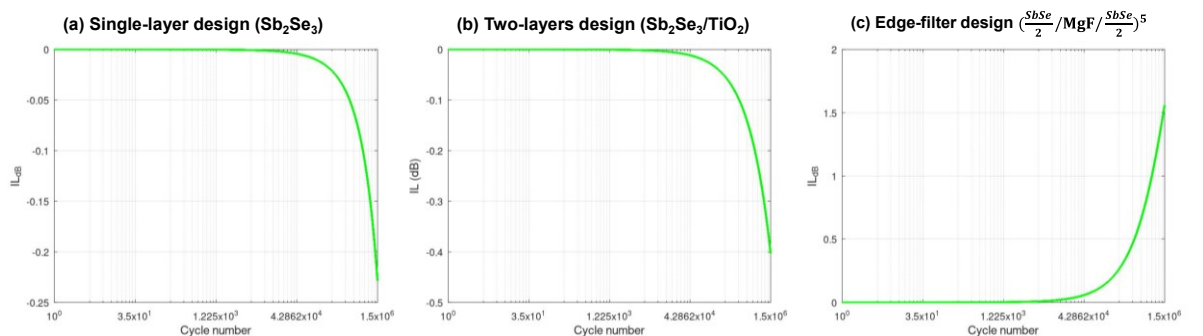


Fig. 11. Insertion loss after 1.5×10^6 phase switching cycles of (a) single-layer, (b) two-layer, and (c) 5-unit-cells edge-filter devices. **Note the different in y-axis scales of the three subfigures.**

Endurance simulations, as shown in figure 9-11, confirm that the devices remain functional for up to 1.5×10^6 phase switching cycles. With gradual but design-dependent degradation of both key figure of merits. Even so, the long-pass edge-filter design still has the best ΔT after enduring more than a million cycles.

4.3 Limitations and future work suggestions

Edge-filter designs have multiple layers of Sb_2Se_3 which complicates the phase switching setup. When laser induced heating is used as a switching method, the phase transition of every Sb_2Se_3 layers would not occur simultaneously, and the transmission will change in a gradual manner. This effect is seen in Sarangan, et al. [3]'s self-activating optical limiter, the device has the same long-pass edge-filter $(\frac{H}{2} L \frac{H}{2})^4$ design but their H layers are GST instead of Sb_2Se_3 , and their L layers are SiO_2 instead of MgF_2 . Their experimental result shows that the phase switching of GST layers happens layer-by-layer in an order that does not necessary started from air-side to substrate-side, but is determined by the changes in optical field distribution and thermal crosstalk between layers. But our device is intended to be use as a switchable filter which does not require a fast switching speed, so electrical heating is viable as an alternative phase switching method. The ability to manually control the phase of each Sb_2Se_3 layer will enable a tunable filter. For the same 5-unit-cells edge-filter design described in Table 1, it consists of 11 layers of Sb_2Se_3 and MgF_2 , 6 layers being Sb_2Se_3 , so there would be a total of 2^6 configuration of Sb_2Se_3 in amorphous or crystalline phases. Figure 12 shows some of the configuration, where the legends denote Sb_2Se_3 in amorphous (a) or crystalline (c) phases, and the ordering from left to right is substrate to air side. The transmittance at 1064 nm can be reduced from 93.4% in ON mode (solid black) in a gradual manner until it is reduced to 0% in OFF mode (dotted black).

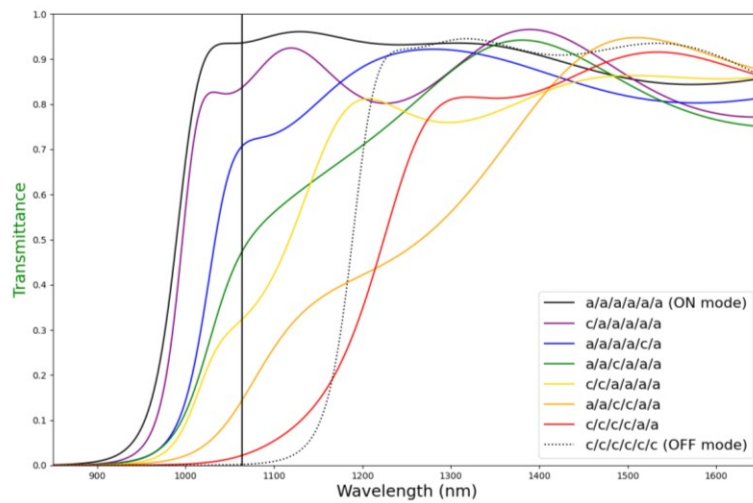
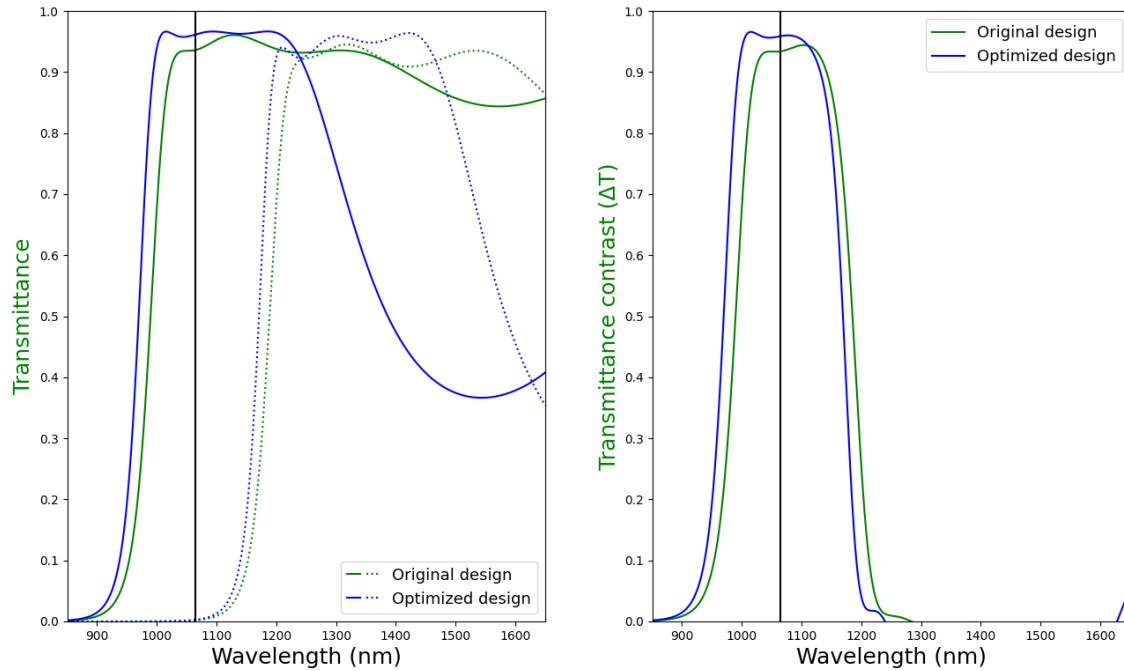


Fig. 12. Transmittance of the same five-unit-cells edge-filter design in 8 different configuration. Each Sb_2Se_3 layer is in amorphous (a) or crystalline (c) phases, and the layouts from left to right are from substrate-side to air-side.

It can clear be seen in Fig. 7 that not all edge-filter designs have the highest ΔT at this wavelength. This can be explained by the selection of reference wavelength ($\lambda_0 = 740$ nm) which is done manually, so an optimization could be used to improve the optical performances. A simple optimization scheme, such as merit function found in OpenFilters [40], can be used to reduce the reflectance in the ON mode by changing the thickness of each layer. For example, we set the optimization goals for the 5-unit-cells edge-filter design to a transmittance of at least 97% in an interval of 20 nm starting from 1000 to 1200 nm. After less than 5 seconds of optimization, the new design now required 304.2 nm of Sb_2Se_3 , 25% more than the originals, but overall thickness is 845.6 nm (93% of the original), the thickness comparison of each layer is shown in Table 2. Figure 13a shows the transmittance of 5-unit-cells edge-filter design after optimization (blue) compared to the original design (green), the optimized design has an overall higher transmittance from 1000 - 1200 nm, as targeted. But the spectral profile of the optimized design has been blue shifted by a small amount, and the transmittances outside the targeted area ($\lambda > 1200$ nm) decrease significantly. Figure 13b is the transmittance contrast of the original and optimized design, ΔT of the optimized design at 1064 nm is now 95.8% (original: 93.4%), and the FWHM has slightly increased to 207.4 nm (original: 197.0 nm).

Table 2. Layers thickness before and after optimization. Sb_2Se_3 and MgF_2 layers are in nm.

Design	SiO_2	Sb_2Se_3	MgF_2	Sb_2Se_3	MgF_2	Sb_2Se_3	MgF_2	Sb_2Se_3	MgF_2	Sb_2Se_3	MgF_2	Sb_2Se_3
Original	3 nm	24.3	134.0	48.5	134.0	48.5	134.0	48.5	134.0	48.5	134.0	24.3
Optimized	3 nm	46.1	67.5	59.3	148.1	55.0	89.3	50.0	145.3	63.3	91.2	30.5

**Fig. 13.** (a) Comparison of transmittance of 5-unit-cells edge-filter designs before (green) and after optimization (blue). The optimized design has an overall higher transmittance in ON mode (solid) starting from around 900 to 1200 nm. (b) ΔT comparison of the two designs, again, the optimized design has an overall higher contrast.

The backside reflection from the substrate-air layer (R_b) creates a slight reflectance. A single layer of MgF_2 deposited on the bottom of the substrate can reduce this effect. This reflection can simply be reduced by applying a translucent adhesive tape to the bottom of the substrate, which create a scattering effect [41].

Our work still lacks thermal analysis and fabrication method, the former would determine the wavelength, intensity, and pulse duration of the laser used for phase switching. Our suggestion for future research would be on fabrication process and experiment on the phase switching method. The phase configuration and thickness optimization of each Sb_2Se_3 layer might also be an interesting topic.

5. Conclusions

We have shown that Sb_2Se_3 can be used as a high-efficient, broadband, tunable optical filter in 1064 nm wavelength. Long-pass edge-filter design has the highest transmission contrast between amorphous and crystalline phases, with 5 unit-cells of Sb_2Se_3 and MgF_2 having the best compromise between optical performances and fabrication difficulties. The design required 11 layers and is 912.6 nm thick, the transmittance contrast at 1064 nm is 93.4% with FWHM ~ 197 nm. After simulating for 1.5×10^6 phase switching cycles, the transmittance contrast degradation and insertion loss of this design are still within acceptable ranges. The use of multiple Sb_2Se_3 layers also allows for tunability, and an optimization of thickness via OpenFilters increase the transmittance contrast at the operating wavelengths of 900 to 1200 nm. This work highlights Sb_2Se_3 as a suitable phase change material for low-loss, high-endurance, tunable transmittance infrared optical filter.

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